

○ *restart;*

○ *edo d y'' C 0.447<sup>2</sup> \$y = 0*

$$\text{edo} := \frac{d^2}{dx^2} y(x) \text{ C } 0.199809 y(x) = 0 \quad (1)$$

○ *cond d y(0) = 1, D(y)(0) = 2;*

$$\text{cond} := y(0) = 1, D(y)(0) = 2 \quad (2)$$

○ *sol d dsolve( {edo, cond}, y(x) );*

$$\text{sol} := y(x) = \frac{2000}{447} \sin\left(\frac{447}{1000} x\right) \text{ C } \cos\left(\frac{447}{1000} x\right) \quad (3)$$

○ *des d rhs(sol);*

$$\text{des} := \frac{2000}{447} \sin\left(\frac{447}{1000} x\right) \text{ C } \cos\left(\frac{447}{1000} x\right) \quad (4)$$

○ *vel d diff(des, x);*

$$\text{vel} := 2 \cos\left(\frac{447}{1000} x\right) \text{ K } \frac{447}{1000} \sin\left(\frac{447}{1000} x\right) \quad (5)$$

○ *acel d diff(des, x\$2);*

$$\text{acel} := \text{K } \frac{447}{500} \sin\left(\frac{447}{1000} x\right) \text{ K } \frac{199809}{1000000} \cos\left(\frac{447}{1000} x\right) \quad (6)$$

○ *ur d 4.585;*

$$\text{ur} := 4.585 \quad (7)$$

○ *vmax d 2.05;*

$$\text{vmax} := 2.05 \quad (8)$$

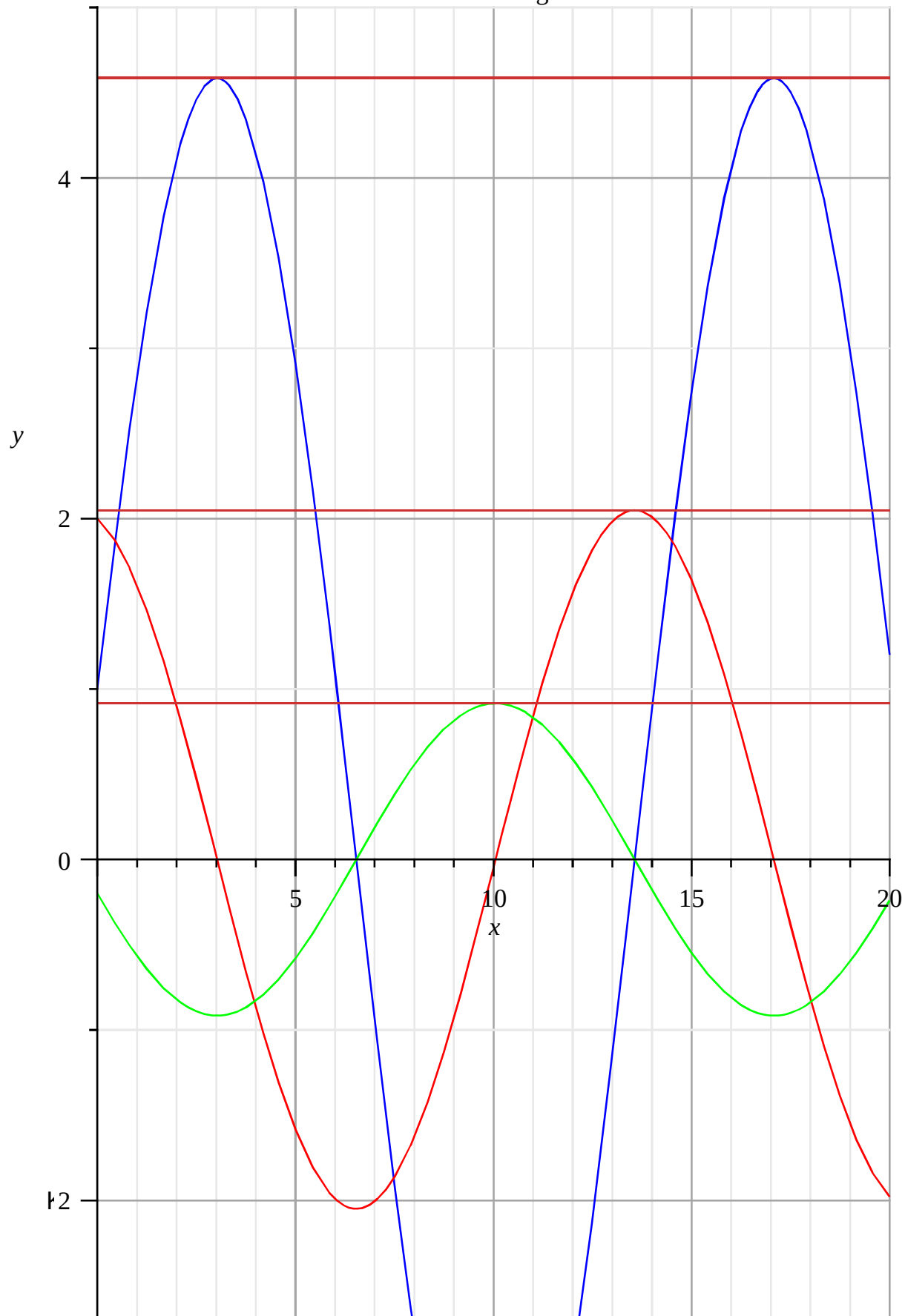
○ *amax d 0.916;*

*amax* := 0.916

(9)

○ `plot( [des, vel, acel, ur, vmax, amax], x = 0 ..20, y =  
15 ..5, color = [blue, red, green, orange, orange,  
orange], legend = ["u(t)", "u'(t)", "u''(t)", "uo",  
"vmax", "amax"], title  
= "Vibración libre sin amortiguamiento", gridlines  
= true );`

# Vibración libre sin amortiguamiento



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